The Number Theory Reference

Organized by rewrite moves, convolution algebra, and sublinear summation templates.

Notation: $[P]$ is 1 if P holds, else 0. $\varepsilon(n) = [n = 1]$, $\mathbf{1}(n) = 1$, $\mathrm{Id}(n) = n$, $\mathrm{Id}_k(n) = n^k$. $\omega(n)$ = number of distinct prime factors, $\Omega(n)$ = prime factors with multiplicity, $\mathrm{rad}(n) = \prod_{p|n} p$. Dirichlet convolution: $(f * g)(n) = \sum_{d|n} f(d)g(n/d)$. Capital letters denote prefix sums: $F(n) = \sum_{i \le n} f(i)$.

0. The Four Moves

Everything else is lookup. These are the edits.

Move	Rule	When
Expand a value	$n \to \sum_{d\|n} \varphi(d)$, $\quad n^k \to \sum_{d\|n} J_k(d)$	a value blocks the sum
Expand a condition	$[n=1] \to \sum_{d\|n} \mu(d)$, $\quad [n \text{ sq}] \to \sum_{d\|n} \lambda(d)$	a condition blocks the sum
Swap the order	$\sum_i \sum_{d\|i} \to \sum_d \sum_{i:\, d\|i}$, inner count $= \lfloor n/d \rfloor$	always, right after expanding
Collect on $T = de$	$\sum_d \sum_e \to \sum_T \lfloor n/T \rfloor (\cdots)\, h(T)$, h a convolution	two nested divisor sums

Plus the two facts that make the swap pay:

$$\#\{i \le n : d \mid i\} = \left\lfloor \frac{n}{d} \right\rfloor \qquad \left\lfloor \frac{\lfloor n/a \rfloor}{b} \right\rfloor = \left\lfloor \frac{n}{ab} \right\rfloor$$

And the **classification lemma**, which generates identities so you stop memorizing them:

$$\#\{i \le n : \gcd(i,n) = d\} = \varphi(n/d) \implies \sum_{i=1}^{n} f(\gcd(i,n)) = \sum_{d|n} f(d)\,\varphi(n/d)$$

1. Multiplicative Function Catalog

A multiplicative f is determined by $f(p^k)$. This table is how you feed `multiplicative_sieve` and how you check whether a function from a problem statement is multiplicative at all.

f	$f(p^k)$	$f(n)$ **closed form**	**Multiplicative?**
ε	$[k=0]$	$[n=1]$	yes (completely)
$\mathbf{1}$	1	1	yes (completely)
Id_k	p^{jk} at p^j (p^k for $k=1$)	n^k	yes (completely)
μ	-1 if $k=1$, 1 if $k=0$, else 0	$(-1)^{\omega(n)}$ if squarefree, else 0	yes
φ	$p^k - p^{k-1}$	$n\prod_{p\|n}(1-1/p)$	yes
$d = \sigma_0$	$k+1$	$\prod(e_i+1)$	yes
$\sigma = \sigma_1$	$\frac{p^{k+1}-1}{p-1}$	$\prod \frac{p_i^{e_i+1}-1}{p_i-1}$	yes
σ_j	$\frac{p^{j(k+1)}-1}{p^j-1}$	$\prod \frac{p_i^{j(e_i+1)}-1}{p_i^j-1}$	yes
J_k (Jordan)	$p^{jk} - p^{(j-1)k}$ at p^j	$n^k\prod_{p\|n}(1-p^{-k})$	yes
ψ (Dedekind)	$p^k + p^{k-1}$	$n\prod_{p\|n}(1+1/p)$	yes
$\mu^2 = \|\mu\|$	$[k \le 1]$	$[n \text{ is squarefree}]$	yes
2^ω	2 (for $k \ge 1$), 1 (for $k=0$)	#squarefree divisors	yes
λ (Liouville)	$(-1)^k$	$(-1)^{\Omega(n)}$	yes (completely)
rad	p (for $k \ge 1$)	$\prod_{p\|n} p$	yes
ω	1 (for $k \ge 1$)		no (additive)
Ω	k		no (completely additive)
Λ (von Mangoldt)	$\log p$ (for $k \ge 1$, else 0)		no

Useful shapes: $d(n)$ is odd iff n is a perfect square. $\sigma(n)$ is odd iff n is a square or twice a square. $\varphi(n)$ is even for $n > 2$. $\sum_{d|n} d = \sigma(n)$, $\prod_{d|n} d = n^{d(n)/2}$.

Jordan and Dedekind functions

$J_k = \mathrm{Id}_k * \mu$ counts k-tuples $(a_1, \ldots, a_k)$ with $\gcd(a_1, \ldots, a_k, n) = 1$ and $1 \le a_i \le n$. Note $J_1 = \varphi$. Moreover, $\sum_{d|n} J_k(d) = n^k$ the direct generalization of $\sum_{d|n} \varphi(d) = n$, which is the "expand a value" rule for n^k.

2. Dirichlet Convolution The Algebra

Commutative, associative, identity ε, distributes over $+$. Multiplicative $*$ multiplicative $=$ multiplicative. Every f with $f(1) \neq 0$ has a Dirichlet inverse.

Distributive rule for completely multiplicative h: $\quad h \cdot (f * g) = (h \cdot f) * (h \cdot g) \quad$ (e.g., $h = \mathrm{Id}_k$)

Convolution	Reads as	Convolution	Reads as
$\mu * \mathbf{1} = \varepsilon$	$\sum_{d\mid n} \mu(d) = [n = 1]$	$\varphi * \sigma = \mathrm{Id} \cdot d$	$\sum_{d\mid n} \varphi(d)\sigma(n/d) = n\, d(n)$
$\varphi * \mathbf{1} = \mathrm{Id}$	$\sum_{d\mid n} \varphi(d) = n$	$\mathrm{Id}_k * \mu = J_k$	$\sum_{d\mid n} J_k(d) = n^k$
$\mu * \mathrm{Id} = \varphi$	$\varphi(n) = \sum_{d\mid n} \mu(d)\frac{n}{d}$	$\mathrm{Id} * \mu^2 = \psi$	Dedekind psi
$\mathbf{1} * \mathbf{1} = d$	$d(n) = \sum_{d\mid n} 1$	$\mathbf{1} * \mu^2 = 2^\omega$	$\sum_{d\mid n} \lvert\mu(d)\rvert = 2^{\omega(n)}$
$\mathrm{Id} * \mathbf{1} = \sigma$	$\sigma(n) = \sum_{d\mid n} d$	$\mathbf{1} * \lambda = [n \text{ square}]$	"expand a square condition"
$\mathrm{Id}_j * \mathbf{1} = \sigma_j$	$\sigma_j(n) = \sum_{d\mid n} d^j$	$\lambda * \lvert\mu\rvert = \varepsilon$	inverse of μ^2 is λ
$\mu * d = \mathbf{1}$	$\sum_{d\mid n} \mu(d)d(n/d) = 1$	$\mathbf{1} * 2^\omega = d(n^2)$	$d(n^2) = \sum_{d\mid n} 2^{\omega(d)}$
$\mu * \sigma = \mathrm{Id}$	$\sum_{d\mid n} \mu(d)\sigma(n/d) = n$	$d * 2^\omega = d^2$	$d(n)^2 = \sum_{d\mid n} d(n/d)2^{\omega(d)}$
$\varphi * d = \sigma$	$\sum_{d\mid n} \varphi(d)d(n/d) = \sigma(n)$	$\mathbf{1} * \Lambda = \log$	$\sum_{d\mid n} \Lambda(d) = \log n \iff \mu * \log = -\Lambda$

Non-convolution and product-divisor identities worth having:

$$\sum_{d\mid n} \frac{\mu^2(d)}{\varphi(d)} = \frac{n}{\varphi(n)} \qquad \sum_{d\mid n} \mu(d)\left(\frac{n}{d}\right)^k = J_k(n) \qquad d(i\cdot j) = \sum_{x\mid i}\sum_{y\mid j}[\gcd(x,y) = 1]$$

Möbius inversion (both directions) & Fast Dirichlet Transforms

- **Divisor form:** $g(n) = \sum_{d\mid n} f(d) \iff f(n) = \sum_{d\mid n} \mu(d)g(n/d)$.
- **Multiple form (used more in CP):** $g(n) = \sum_{n\mid m} f(m) \iff f(n) = \sum_{n\mid m} \mu(m/n)g(m)$.

In code, prefer the subtract-downward loops over writing μ by hand you cannot get a sign wrong:

```
// Divisible-by -> Exactly (O(N log N), Mobius over multiples)
for (int d = N; d >= 1; d--) for (int k = 2*d; k <= N; k += d) cnt[d] -= cnt[k];
// Exactly -> Divisible-by
for (int d = 1; d <= N; d++) for (int k = 2*d; k <= N; k += d) cnt[d] += cnt[k];

// Fast Dirichlet Transforms in O(N log log N) when one side is 1 or mu:
for (int p : primes) for (int i = 1; (ll)i*p <= N; i++) f[i*p] += f[i]; // f = f * 1 (divisor zeta)
for (int p : primes) for (int i = N/p; i >= 1; i--)     f[i*p] -= f[i]; // f = f * mu (divisor
    mobius)
for (int p : primes) for (int i = N/p; i >= 1; i--)     f[i] += f[i*p]; // multiple zeta
for (int p : primes) for (int i = 1; (ll)i*p <= N; i++) f[i] -= f[i*p]; // multiple mobius
```

3. Dirichlet Series How to Find the Right Convolution

This lookup table tells you what to pair with what. Read a row, factor the right side into things you can prefix-sum, and you have your Du-sieve pairing. Key tags: $\zeta(s) \leftrightarrow \mathbf{1}$, $\zeta(s-k) \leftrightarrow \mathrm{Id}_k$, $1/\zeta(s) \leftrightarrow \mu$, $\zeta(2s) \leftrightarrow [n \text{ is a square}]$.

$f(n)$	$\sum_n f(n)n^{-s}$	$f(n)$	$\sum_n f(n)n^{-s}$	$f(n)$	$\sum_n f(n)n^{-s}$
$\mathbf{1}$	$\zeta(s)$	J_k	$\zeta(s-k)/\zeta(s)$	$d(n^2)$	$\zeta(s)^3/\zeta(2s)$
d	$\zeta(s)^2$	μ^2	$\zeta(s)/\zeta(2s)$	$d(n)^2$	$\zeta(s)^4/\zeta(2s)$
σ_k	$\zeta(s)\zeta(s-k)$	2^ω	$\zeta(s)^2/\zeta(2s)$	Λ	$-\zeta'(s)/\zeta(s)$
μ	$1/\zeta(s)$	λ	$\zeta(2s)/\zeta(s)$	$\mathrm{Id}\cdot\varphi$	$\zeta(s-2)/\zeta(s-1)$
φ	$\zeta(s-1)/\zeta(s)$	ψ	$\zeta(s)\zeta(s-1)/\zeta(2s)$	$\mathrm{Id}_k\cdot\varphi$	$\zeta(s-k-1)/\zeta(s-k)$

Worked use: Want $\sum_{i\le n} \mu^2(i)$? The row says $\zeta(s)/\zeta(2s)$, i.e., $\mu^2 = \mathbf{1} * (\mu \text{ on squares})$, so $\mu^2(n) = \sum_{d^2\mid n} \mu(d)$. Swap the order and you get the $O(\sqrt{n})$ closed form in §5 with no recurrence at all.

4. $\lfloor n/i \rfloor$ Block Decomposition & Hyperbola Method

$\lfloor n/i \rfloor$ takes $O(\sqrt{n})$ distinct values in contiguous blocks. Given l, the block ends at $r = \lfloor n/\lfloor n/l \rfloor \rfloor$.

```
for (ll l = 1, r; l <= n; l = r + 1) { r = n / (n / l); /* q = n/l is constant on [l, r] */ }
// two args: r = min(n / (n / l), m / (m / l)); iterate l <= min(n, m)
```

The Master Pattern: $\sum_i f(i)\, g(\lfloor n/i \rfloor)$ with F prefix-summable $\Rightarrow O(\sqrt{n})$:

$$\sum_{i=1}^{n} f(i)\, g\Big(\Big\lfloor \frac{n}{i} \Big\rfloor\Big) = \sum_{\text{blocks } [l,r]} \big(F(r) - F(l-1)\big)\, g\Big(\Big\lfloor \frac{n}{l} \Big\rfloor\Big)$$

The $T = de$ Collection: Given $\sum_d w(d) \sum_e \mu(e) \lfloor \frac{n}{de} \rfloor \lfloor \frac{m}{de} \rfloor$, set $T = de$:

$$= \sum_{T} \Big\lfloor \frac{n}{T} \Big\rfloor \Big\lfloor \frac{m}{T} \Big\rfloor h(T), \qquad h = w * \mu$$

Sieve h once ($O(L \log\log L)$ via fast divisor Möbius), prefix-sum it, then every query is one block loop.

Dirichlet Hyperbola Method: For $\sum_{i \le n} (f * g)(i)$ when both F and G are easy to compute, in $O(\sqrt{n})$:

$$\sum_{i \le n} (f*g)(i) = \sum_{a \le \sqrt{n}} f(a) G\Big(\Big\lfloor \frac{n}{a} \Big\rfloor\Big) + \sum_{b \le \sqrt{n}} g(b) F\Big(\Big\lfloor \frac{n}{b} \Big\rfloor\Big) - F(\lfloor \sqrt{n} \rfloor) G(\lfloor \sqrt{n} \rfloor)$$

`floor_sum`, $O(\log m)$: $\sum_{i=0}^{n-1} \lfloor \frac{ai+b}{m} \rfloor$ Euclid-like recursion for lattice points under a line. Use when the numerator is linear in i, not a constant divided by i.

5. Prefix Sums at Huge n The Du Sieve Table

The Recipe: Pick g with an easy prefix sum G such that $h = f * g$ also has an easy prefix sum H. Then from $H(n) = \sum_{d \le n} g(d) F(\lfloor n/d \rfloor)$ and $g(1) = 1$:

$$\boxed{F(n) = H(n) - \sum_{d=2}^{n} g(d)\, F\Big(\Big\lfloor \frac{n}{d} \Big\rfloor\Big)}$$

Block the right side, memoize, and linear-sieve F up to $\approx n^{2/3}$ as the base case. Time complexity: $O(n^{2/3})$.

f	g	$h = f * g$	$H(n) = \sum_{i \le n} h(i)$	$G(n) = \sum_{i \le n} g(i)$
φ	$\mathbf{1}$	Id	$\frac{n(n+1)}{2}$	n
μ	$\mathbf{1}$	ε	1	n
$\mathrm{Id} \cdot \varphi$	Id	Id_2	$\frac{n(n+1)(2n+1)}{6}$	$\frac{n(n+1)}{2}$
$\mathrm{Id} \cdot \mu$	Id	ε	1	$\frac{n(n+1)}{2}$
$\mathrm{Id}_k \cdot \varphi$	Id_k	Id_{k+1}	$\sum_{i \le n} i^{k+1}$ (Faulhaber)	$\sum_{i \le n} i^k$
J_k	$\mathbf{1}$	Id_k	$\sum_{i \le n} i^k$ (Faulhaber)	n
λ	$\mathbf{1}$	[square]	$\lfloor \sqrt{n} \rfloor$	n

The two you'll write most:

$$\Phi(n) = \frac{n(n+1)}{2} - \sum_{k=2}^{n} \Phi\Big(\Big\lfloor \frac{n}{k} \Big\rfloor\Big) \qquad M(n) = 1 - \sum_{k=2}^{n} M\Big(\Big\lfloor \frac{n}{k} \Big\rfloor\Big)$$

The ones that need no recurrence $O(\sqrt{n})$ outright

$$\sum_{i \le n} d(i) = \sum_{i \le n} \Big\lfloor \frac{n}{i} \Big\rfloor = 2 \sum_{i \le \sqrt{n}} \Big\lfloor \frac{n}{i} \Big\rfloor - \lfloor \sqrt{n} \rfloor^2 \qquad \sum_{i \le n} \sigma(i) = \sum_{i \le n} i \Big\lfloor \frac{n}{i} \Big\rfloor \text{ (block it)}$$

$$\boxed{\sum_{i \le n} \mu^2(i) = \#\{\text{squarefree} \le n\} = \sum_{d \le \sqrt{n}} \mu(d) \Big\lfloor \frac{n}{d^2} \Big\rfloor} \qquad \#\{k\text{-free} \le n\} = \sum_{d \le n^{1/k}} \mu(d) \Big\lfloor \frac{n}{d^k} \Big\rfloor$$

$$\sum_{i \le n} 2^{\omega(i)} = \sum_{d \le n} \mu^2(d) \Big\lfloor \frac{n}{d} \Big\rfloor \text{ (block using } \mu^2 \text{ prefix)}$$

Beyond Du: when f has no nice g

- **Powerful-Number Trick:** If $f(p) = g(p)$ on primes for a simple multiplicative g, write $f = g * h$. Then $h(p) = 0$, so h is nonzero only on *powerful numbers* ($p \mid k \Rightarrow p^2 \mid k$, only $O(\sqrt{n})$ of them $\le n$). Compute $h(p^k)$ via $h(p^k) = f(p^k) - \sum_{j=1}^{k} g(p^j)h(p^{k-j})$, then DFS over powerful numbers $k \le n$ to accumulate $F(n) = \sum_{\text{powerful } k \le n} h(k)\, G(\lfloor n/k \rfloor)$ in $O(\sqrt{n})$.
- **Min_25 Sieve:** Computes $\sum_{p \le n} f(p)$ over primes and full multiplicative prefix sums in $O(n^{3/4}/\log n)$ (also $\pi(n)$ and $\sum_{p \le n} p$ for $n \le 10^{13}$). Reach for it only when prime sums at that scale are required.

6. GCD, LCM, and Divisor Sums

Let $L = \min(n, m)$.

$$\sum_{i \le n} \sum_{j \le m} \gcd(i,j) = \sum_{d \le L} \varphi(d) \left\lfloor \frac{n}{d} \right\rfloor \left\lfloor \frac{m}{d} \right\rfloor$$

$$\#\{(i,j) : \gcd(i,j) = 1\} = \sum_{d \le L} \mu(d) \left\lfloor \frac{n}{d} \right\rfloor \left\lfloor \frac{m}{d} \right\rfloor \;=\; 2\Phi(L) - 1 \text{ when } n = m$$

$$\#\{(i,j) : \gcd(i,j) = k\} = \sum_{e \le \lfloor L/k \rfloor} \mu(e) \left\lfloor \frac{n}{ke} \right\rfloor \left\lfloor \frac{m}{ke} \right\rfloor$$

$$\sum_{i \le n} \gcd(i,n) = \sum_{d \mid n} d\, \varphi(n/d) = \sum_{d \mid n} \varphi(d) \frac{n}{d} \quad \text{(Pillai's function)}$$

$$\sum_{i \le n} \operatorname{lcm}(i,n) = \frac{n}{2}\Big(1 + \sum_{d \mid n} d\, \varphi(d)\Big)$$

$$\sum_{a_1, \ldots, a_r \le M} [\gcd = 1] = \sum_{d \le M} \mu(d) \left\lfloor \frac{M}{d} \right\rfloor^r$$

$$\sum_{i \le n} \sum_{j \le m} \operatorname{lcm}(i,j) = \sum_{T \le L} \Big(\sum_{d \mid T} d\, \mu(d) \Big) S\Big(\Big\lfloor \frac{n}{T} \Big\rfloor\Big) S\Big(\Big\lfloor \frac{m}{T} \Big\rfloor\Big) \cdot T, \quad S(x) = \frac{x(x+1)}{2}$$

$$\sum_{i \le n} \sum_{j \le m} d(i \cdot j) = \sum_{g \le L} \mu(g)\, D\Big(\Big\lfloor \frac{n}{g} \Big\rfloor\Big)\, D\Big(\Big\lfloor \frac{m}{g} \Big\rfloor\Big), \quad D(x) = \sum_{k=1}^{x} \Big\lfloor \frac{x}{k} \Big\rfloor$$

Array Pair GCD Template: Given array $a_1, \ldots, a_n \le M$, let $c(d) = \#\{i : d \mid a_i\}$. Then:

$$\sum_{i,j} w(\gcd(a_i, a_j)) = \sum_{d=1}^{M} (w * \mu)(d)\, c(d)^2$$

Fixed point that costs WA: $\sum_{k \le l,\, \gcd(k,l)=1} k = \frac{l\varphi(l)}{2}$ holds **only for** $l > 1$; at $l = 1$ it is 1.

Necklaces (Burnside $\times$ classification lemma):

$$\text{Total } k\text{-ary necklaces of length } n\text{: } \frac{1}{n} \sum_{d \mid n} \varphi(n/d)\, k^d \qquad \text{With } m \text{ ones: } \frac{1}{n} \sum_{d \mid \gcd(n,m)} \varphi(d) \binom{n/d}{m/d}$$

7. Congruences and Modular Machinery

Tool	Statement	Cost
Fermat / Euler	$a^{p-1} \equiv 1 \pmod p$ $(p \nmid a)$; $a^{\varphi(m)} \equiv 1 \pmod m$ $(\gcd(a,m)=1)$	
Modular Inverse	`egcd` needs only $\gcd(a,m)=1$, not m prime	$O(\log m)$
Linear Inverses	$\text{inv}[i] = -\lfloor m/i \rfloor \cdot \text{inv}[m \bmod i] \bmod m$	$O(n)$
Batch Inverses	prefix products, one `egcd` at end, walk backward	$O(n+\log m)$
Linear Congruence	$ax \equiv b \pmod m$ solvable iff $g=\gcd(a,m) \mid b$ (g sols mod m)	$O(\log m)$
CRT (2 moduli)	solvable iff $g=\gcd(m_1,m_2) \mid (r_2-r_1)$; unique mod $\text{lcm}(m_1,m_2)$	$O(\log m)$
Wilson's Theorem	$(p-1)! \equiv -1 \pmod p$ iff p is prime	
Legendre's Formula	$v_p(n!) = \sum_{k\geq 1}\lfloor n/p^k \rfloor = \frac{n-s_p(n)}{p-1}$ (s_p = sum of base-p digits)	$O(\log_p n)$
Kummer's Theorem	$v_p\binom{n}{r}$ = number of carries when adding $r+(n-r)$ in base p	$O(\log_p n)$
Lucas's Theorem	$\binom{n}{r} \equiv \prod \binom{n_i}{r_i} \pmod p$; $\binom{n}{r}$ odd $\iff (r \,\&\, n) = r$	$O(\log_p n)$
Euler Tower Reduction	$a^e \equiv a^{(e \bmod \varphi(m))+\varphi(m)} \pmod m$ for all $e \geq \log_2 m$ (any a)	$O(\log m)$
Frobenius Coin	$\gcd(a,b)=1 \Rightarrow$ max unreachable is $ab-a-b$; $\frac{(a-1)(b-1)}{2}$ unreachable	$O(1)$
Miller–Rabin	bases $\{2,3,5,7,11,13,17,19,23,29,31,37\}$ deterministic for $n < 2^{64}$	$O(12\log n)$
Pollard's Rho (Brent)	randomized expected smallest factor extraction	$O(n^{1/4})$
BSGS / Tonelli–Shanks	discrete log $a^x \equiv b$ in $O(\sqrt{m})$; modular $\sqrt{a} \bmod p$ in $O(\log^2 p)$	
Hensel's Lemma	lift root $f(r) \equiv 0 \pmod{p^k}$ to p^{k+1} if $f'(r) \not\equiv 0 \pmod p$	$O(k)$

Lifting the Exponent (LTE) Complete Cases ($p \nmid ab$):

- **Odd prime p:** If $p \mid (a-b)$, then $v_p(a^n-b^n) = v_p(a-b)+v_p(n)$. If $p \mid (a+b)$ and n is odd, $v_p(a^n+b^n) = v_p(a+b)+v_p(n)$.
- **For $p=2$ (with a,b odd, $2 \mid a-b$):** If n is odd, $v_2(a^n-b^n) = v_2(a-b)$. If n is even, $v_2(a^n-b^n) = v_2(a-b)+v_2(a+b)+v_2(n)-1$.

Multiplicative Order, Primitive Roots & Discrete Roots: $\text{ord}_n(a) \mid \varphi(n)$. To find it: start at $d=\varphi(n)$ and divide out each prime factor of $\varphi(n)$ while $a^{d/p} \equiv 1 \pmod n$. A primitive root exists iff $n \in \{1,2,4,p^k,2p^k\}$ (there are $\varphi(\varphi(n))$ of them). For prime p, $x^k \equiv a \pmod p$ $(a \not\equiv 0)$ has a solution iff $a^{(p-1)/\gcd(k,p-1)} \equiv 1 \pmod p$.

Quadratic Residues: Legendre symbol $\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod p$. Exactly $\frac{p-1}{2}$ nonzero QRs mod p. -1 is a QR iff $p \equiv 1 \pmod 4$; 2 is a QR iff $p \equiv \pm 1 \pmod 8$. Law of Quadratic Reciprocity: $\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\frac{q-1}{2}}$.

8. Sieves, Costs, and Asymptotics

Sieve	Gives	Cost	Sieve	Gives	Cost
Linear	spf, μ, φ, primes	$O(N)$	Segmented	primes in $[L,R]$	$O((R-L)\log\log R+\sqrt{R})$
Linear mult.	any f from $f(p^k)$	$O(N)$	Du Sieve	$\Phi(n), M(n)$ $(n \leq 10^{10})$	$O(n^{2/3})$
Divisor loop	d, σ, any $\sum_{d\mid n}$	$O(N\log N)$	Min_25	prime / mult. prefix	$O(n^{3/4}/\log n)$

Asymptotics for sizing: $\sum_{i\leq n} d(i) \approx n\ln n+(2\gamma-1)n$; $\sum_{i\leq n}\sigma(i) \approx \frac{\pi^2}{12}n^2$; $\Phi(n) \approx \frac{3}{\pi^2}n^2$; $\#\{\text{squarefree} \leq n\} \approx \frac{6}{\pi^2}n$; $\pi(n) \approx n/\ln n$; $\sum_{p\leq n} 1/p \approx \ln\ln n + M$. Max divisors $\max_{i\leq n} d(i)$: 240 at 10^5, 1344 at 10^7, 6720 at 10^9, 103680 at 10^{18}. Max distinct primes $\omega(n) \leq 15$ for $n \leq 10^{18}$.

The Workflow

1. **Write the brute force as nested sums.** Don't be clever yet.
2. **Expand whatever blocks you** a value with φ or J_k, a condition with μ or λ.
3. **Swap the order of summation.** Inner sum becomes $\lfloor n/d \rfloor$.
4. **If two nested divisor sums remain, collect on $T=de$** and identify the coefficient as a convolution. Sieve it once.
5. **Size against n:** $n \leq 10^7 \to$ sieve and loop. $n \leq 10^{10} \to$ block + Du sieve. Beyond $\to$ Min_25 or Powerful-Number DFS.
6. **Prefer the subtract-downward loop** over writing μ by hand whenever you need all d at once.

7. **Check the modulus** before writing `pw(a, MOD-2)` or dividing by n in Burnside.

8. **Verify against brute force on small n.** Every closed form, every time watch for pairing fixed points ($l = 1, n = 1$).